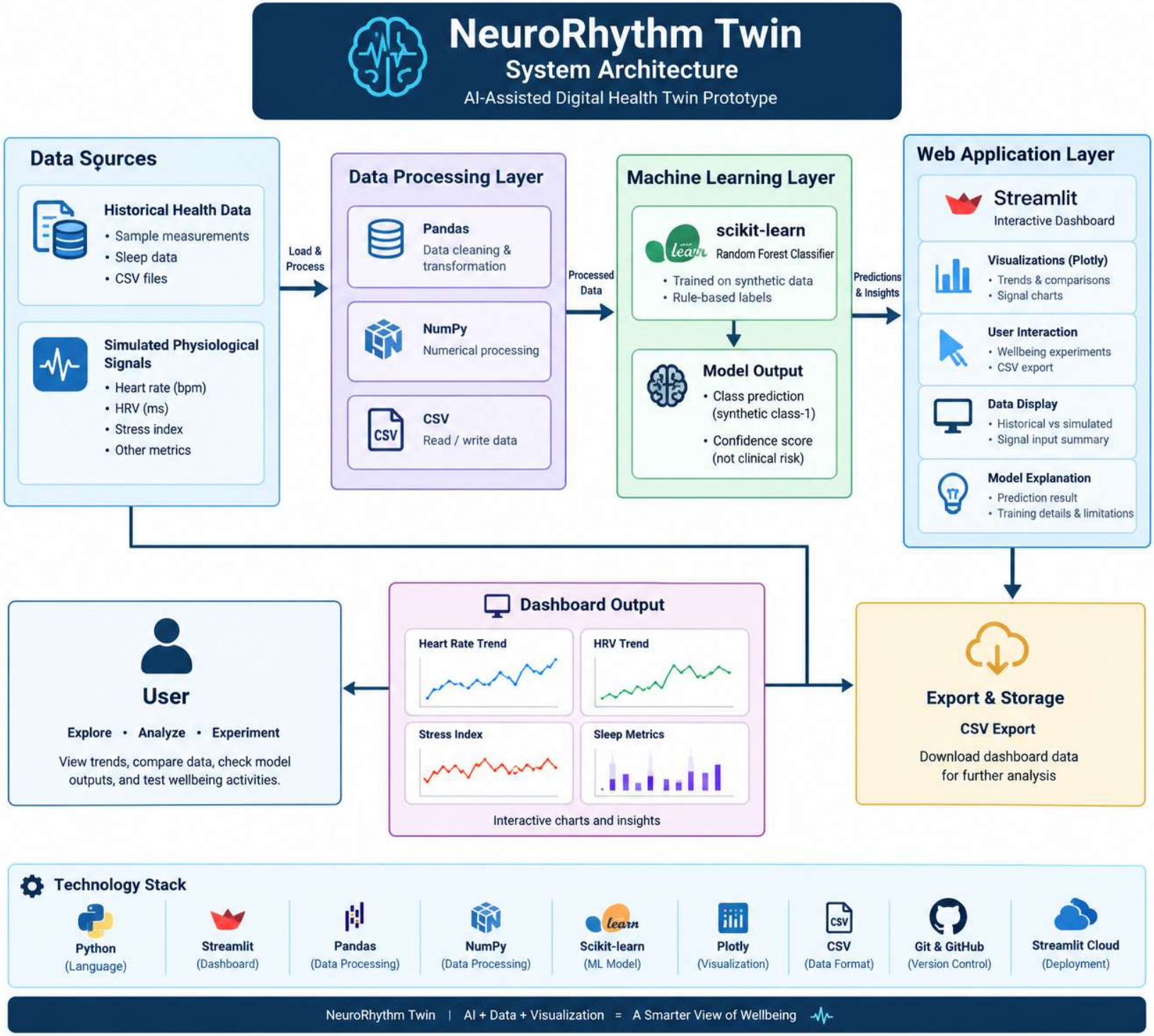

NeuroRhythm Twin
System Architecture
AI-Assisted Digital Health Twin Prototype
Data Sources
Historical Health Data
• Sample measurements
• Sleep data
• CSV files
Simulated Physiological Signals
• Heart rate (bpm)
• HRV (ms)
• Stress index
• Other metrics
Load & Process
Data Processing Layer
Pandas
Data cleaning & transformation
NumPy
Numerical processing
CSV
Read / write data
Processed Data
Machine Learning Layer
scikit-learn
Random Forest Classifier
• Trained on synthetic data
• Rule-based labels
Model Output
• Class prediction (synthetic class-1)
• Confidence score (not clinical risk)
Predictions & Insights
Web Application Layer
Streamlit
Interactive Dashboard
Visualizations (Plotly)
• Trends & comparisons
• Signal charts
User Interaction
• Wellbeing experiments
• CSV export
Data Display
• Historical vs simulated
• Signal input summary
Model Explanation
• Prediction result
• Training details & limitations
User
Explore • Analyze • Experiment
View trends, compare data, check model outputs, and test wellbeing activities.
Dashboard Output
Heart Rate Trend
HRV Trend
Stress Index
Sleep Metrics
Interactive charts and insights
Export & Storage
CSV Export
Download dashboard data for further analysis
Technology Stack
Python (Language)
Streamlit (Dashboard)
Pandas (Data Processing)
NumPy (Data Processing)
Scikit-learn (ML Model)
Plotly (Visualization)
CSV (Data Format)
Git & GitHub (Version Control)
Streamlit Cloud (Deployment)
NeuroRhythm Twin | AI + Data + Visualization = A Smarter View of Wellbeing